5th Experiment: Band Structure Simulation for Graphene and CNTs using NanoHub

Fig 1: This figure illustrates the corresponding simulation setup, model, or output in NanoHub. It helps visualize the device structure, selected physics, or simulation results for better understanding.

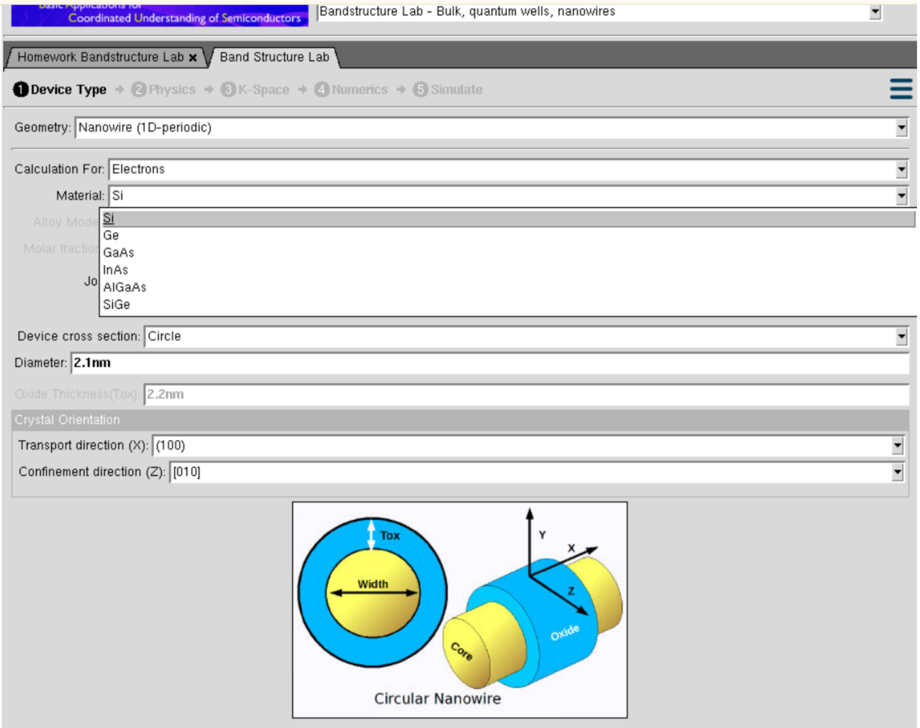


Fig 2: This figure illustrates the corresponding simulation setup, model, or output in NanoHub. It helps visualize the device structure, selected physics, or simulation results for better understanding.

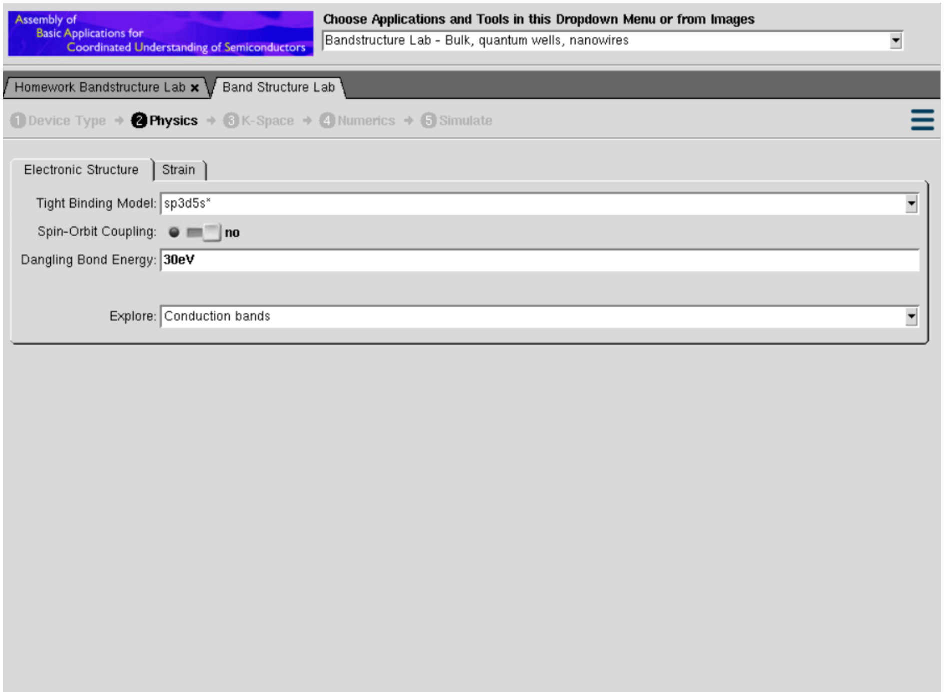
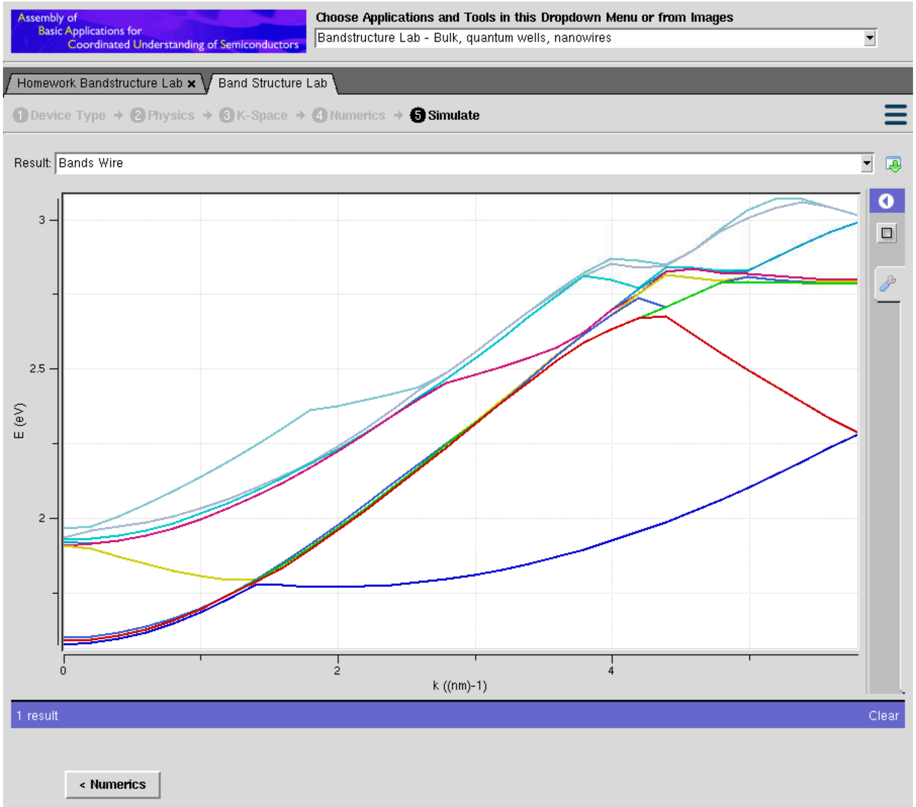


Fig 3: The colored curves represent the variation of electron energy with the wave vector (k), helping to identify the band gap and energy levels.



Case Study 1: This section presents the experiment or case study and its simulation objective. It demonstrates the behaviour and analysis of nanoscale devices using NanoHub.

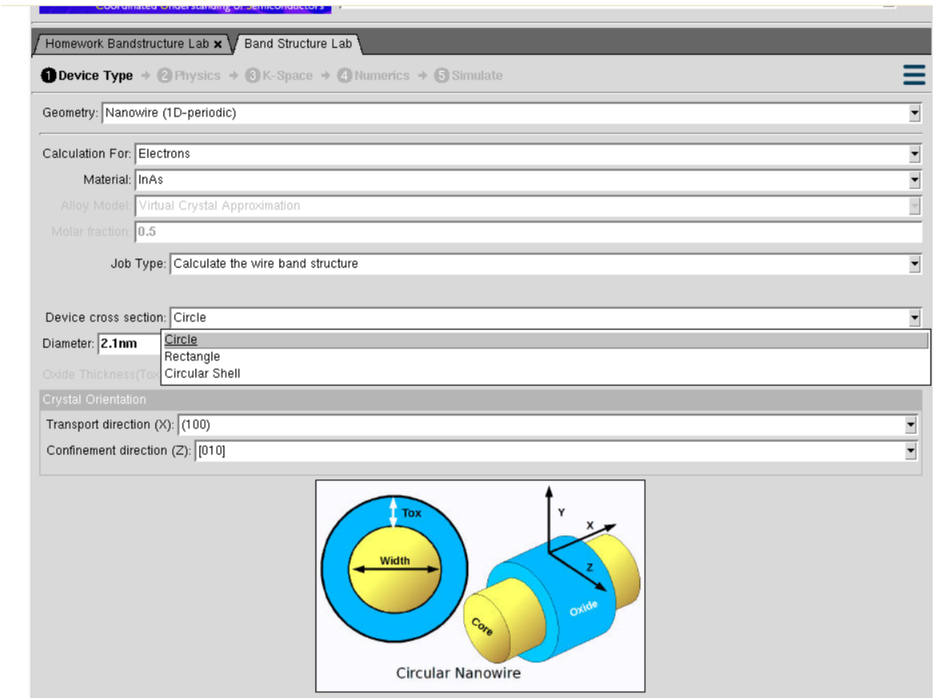
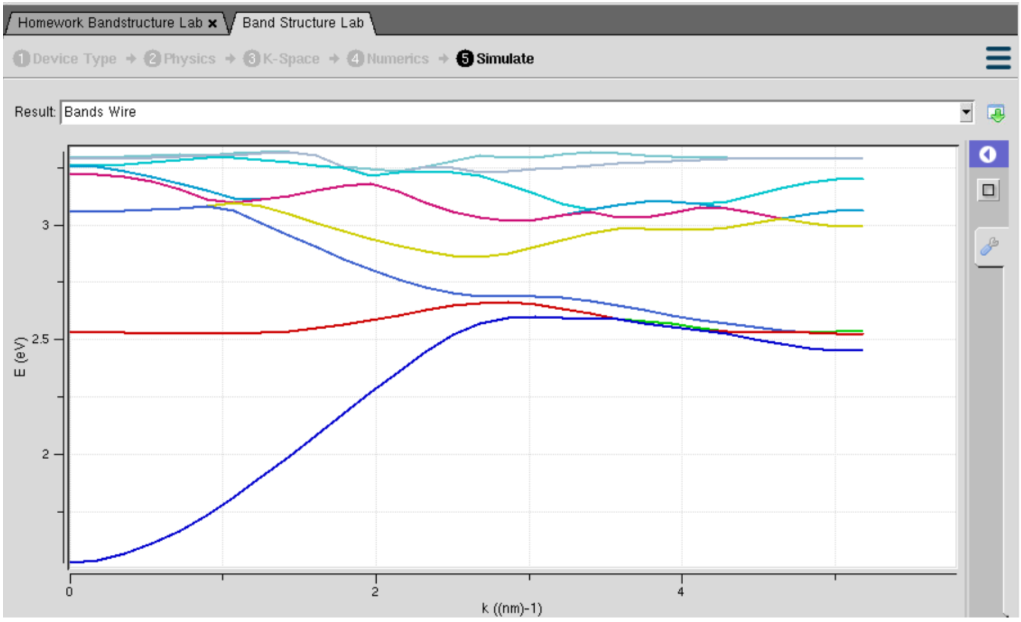


Fig 4: This figure shows the band structure of a semiconductor nanowire simulated using the NanoHub Band Structure Lab. The colored energy bands indicate how electron energy changes with the wave vector (k), revealing the electronic behavior of the material.



Case Study 2: This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

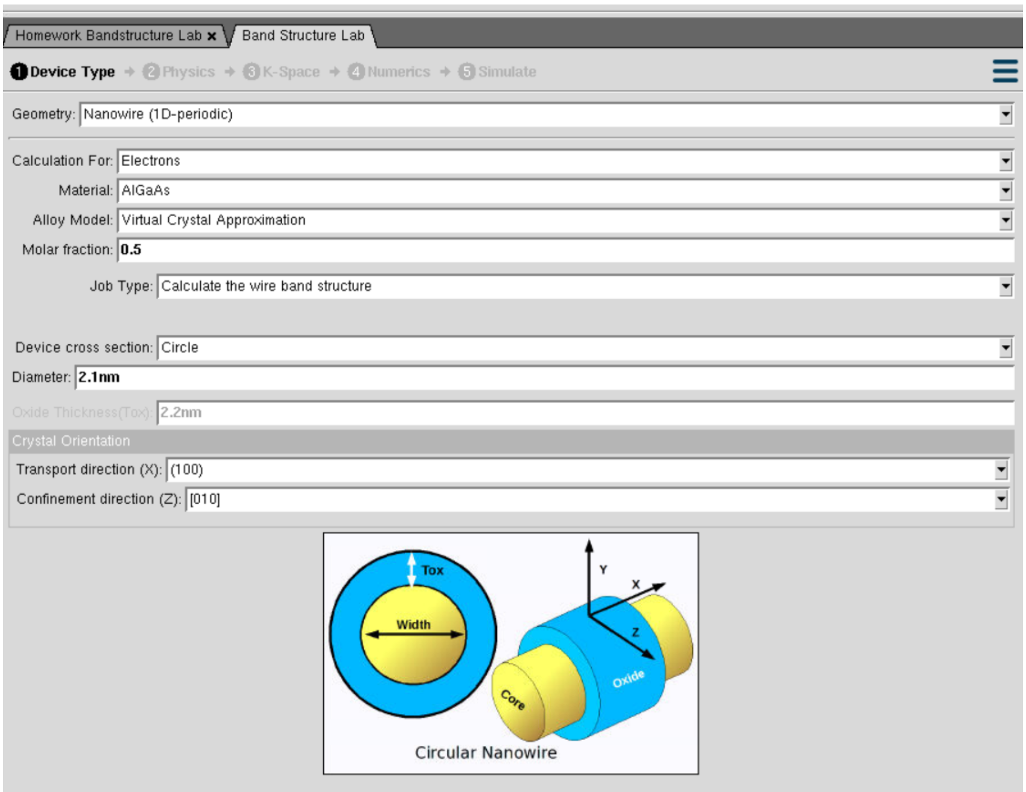
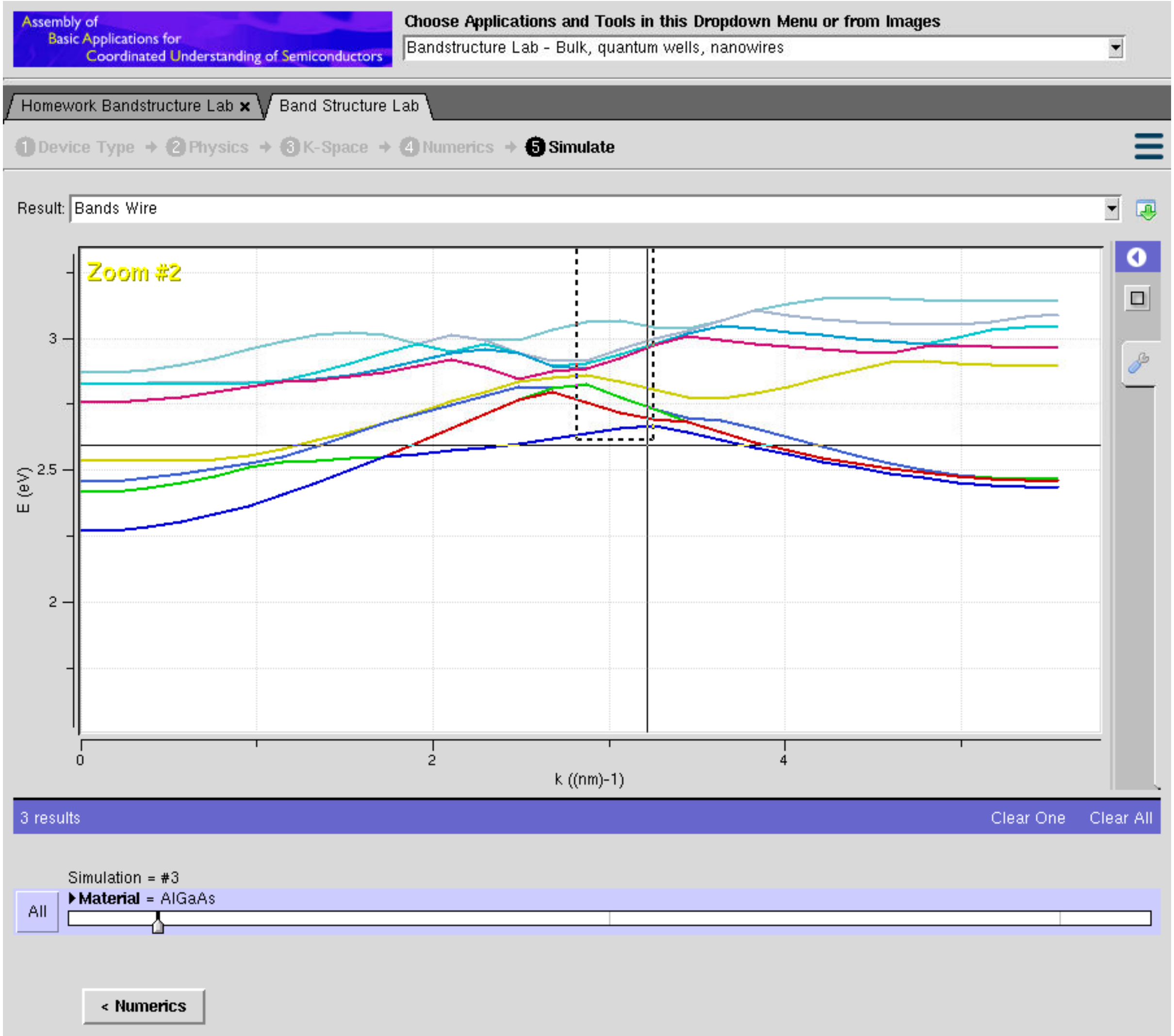
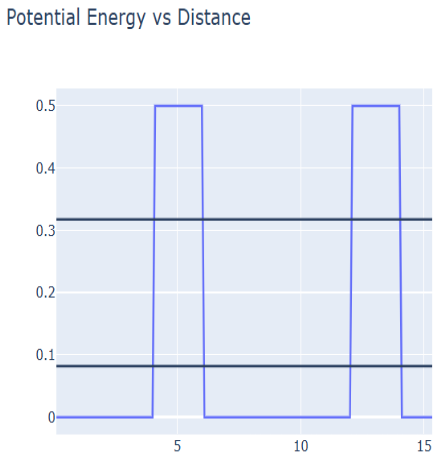


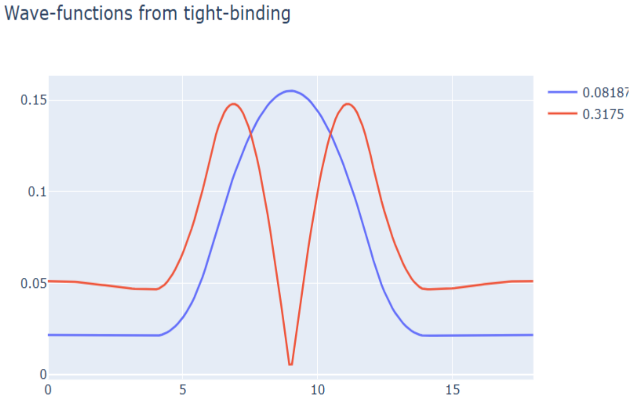
Fig 5: The colored curves represent the allowed energy bands, illustrating how electron energy varies with the wave vector (k).



6th Experiment: Demonstration of Quantum Tunneling

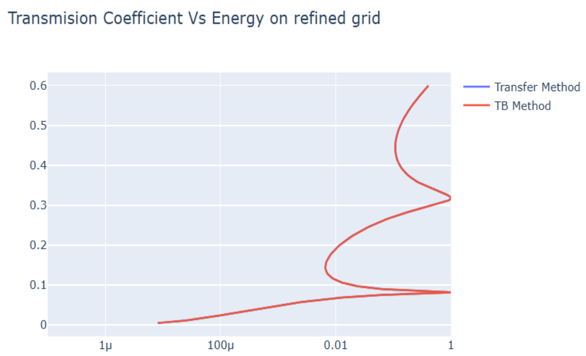
This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub

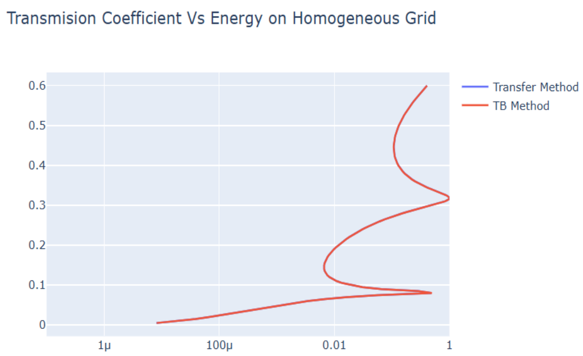




Case Study 1: Analysis the transmission co-efficient Vs Energy characteristics on refined grid

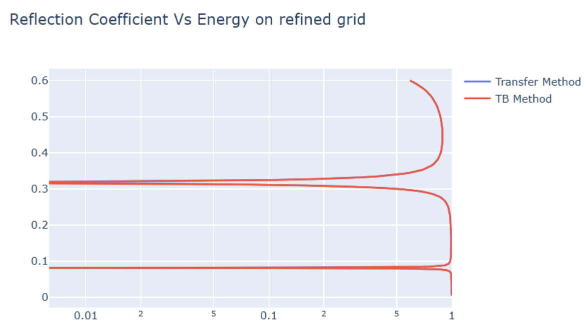
This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

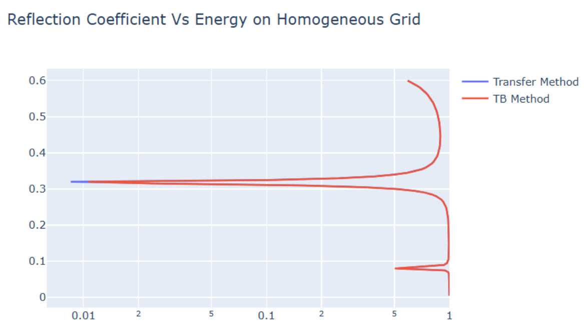




Case Study 2: Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

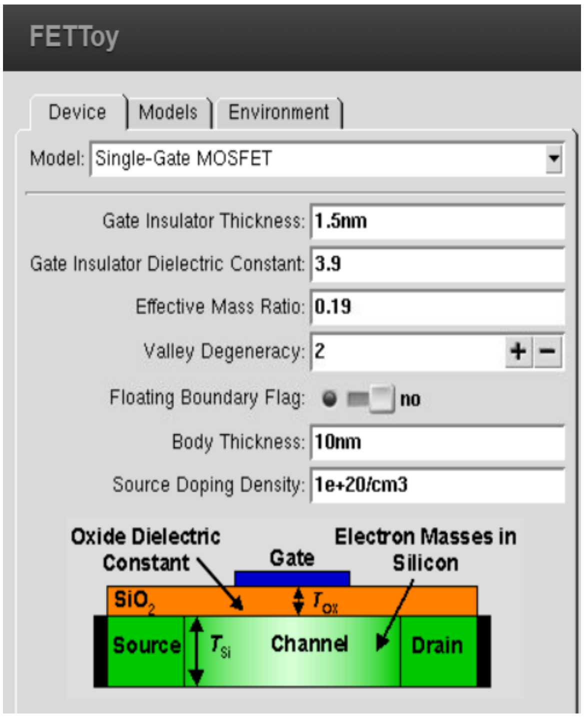
This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

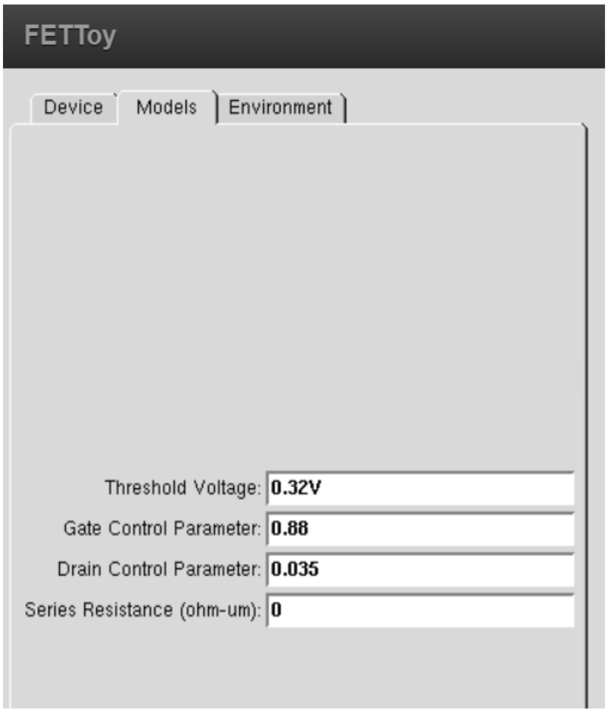




Experiment 7: MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub.

This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.





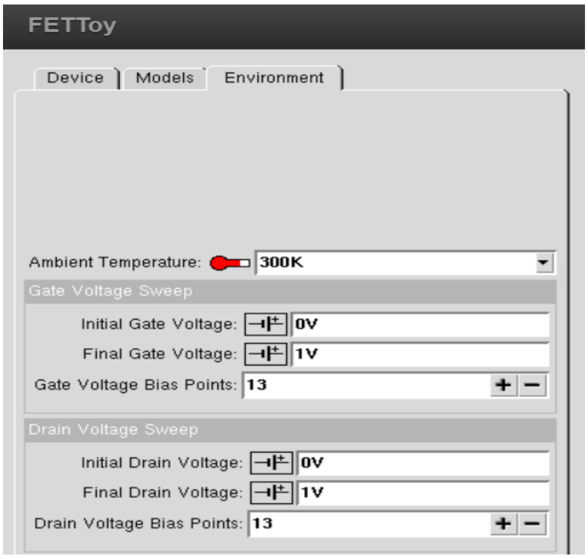


Fig 6: The graph shows that the **drain current increases as the gate voltage increases**, indicating normal MOSFET switching behavior. The **red curve** has a higher current than the **blue curve**, showing better performance.

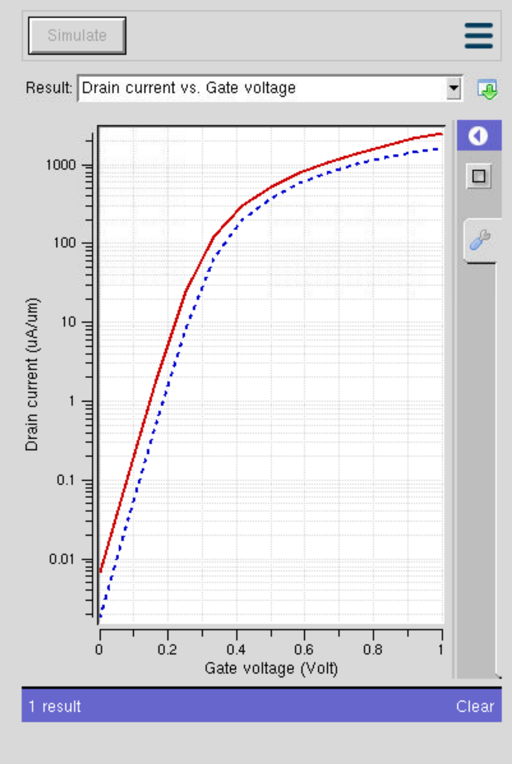


Fig 7: The graph shows that the **drain current increases with drain voltage and then becomes constant (saturation)**. Higher **gate voltage** results in **higher drain current**, indicating normal MOSFET operation.

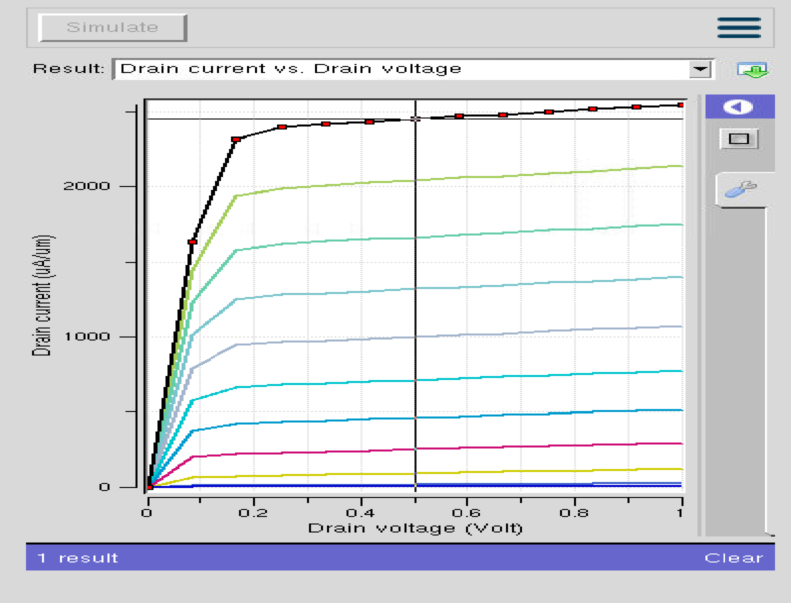


Fig 8: The **gm/Id ratio decreases as the gate voltage increases**. A **higher gm/Id at low gate voltage** indicates better transistor efficiency, while a **lower gm/Id at high gate voltage** indicates stronger inversion operation.